Gyuyeong Kim, Ph.D.

Assistant Professor
Department of Computer Engineering
Sungshin Women's University
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https://gyuyeongkim.github.io/

RESEARCH INTERESTS

Networked systems, Network stack, In-network computing

EDUCATION

Ph.D. in Computer Science, **Korea University**, South Korea — Mar. 2012 - Feb. 2020
- Dissertation: Buffer Sharing Mechanisms for Data Center Networks
- Advisor: Prof. Wonjun Lee

B.S. in Computer Science, **Korea University**, South Korea — Mar. 2008 - Feb. 2012
- Interdisciplinary major: Brain and Cognitive Sciences

Exchange Student, **Lund University**, Sweden — Jan. 2011 - Jun. 2011
- Faculty of Engineering (Lund Tekniska Högskola, LTH)

EMPLOYMENT HISTORY

Assistant Professor, Dept. of Computer Eng., Sungshin Women's University — Mar. 2022 - Present
Research Professor, Future Network Center, Korea University — Mar. 2020 - Mar. 2022
Lecturer, Department of Information Security, Seoul Women's University — Mar. 2021 - Aug. 2022
Lecturer, School of Cybersecurity, Korea University — Sept. 2019 - Feb. 2020

GRANTS

1. 분산 데이터저장소를 위한 프로그래머블 인-네트워크 컴퓨팅 기술, Programmable In-Network Computing for Distributed Data Stores, Basic Research Program (**기본연구**), NRF, Grant: 68,054,000 KRW — Jun. 2023 - May 2024
2. 가변형 네트워크 디바이스를 활용한 인네트워크 자원 증강 기술, In-Network Resource Augmentation utilizing Reconfigurable Network Devices, Young Researcher Program (**우수신진연구**), NRF, Grant: 294,000,000 KRW — Mar. 2020 - Feb. 2023

CONFERENCE PUBLICATIONS

1. Gyuyeong Kim, "NetClone: Fast, Dynamic Request Cloning for Microsecond-Scale RPCs," in *Proc. of 37th Annual Conference of the ACM Special Interest Group on Data Communication* **(SIGCOMM 2023)**, New York, NY, USA, September 2023. (Acceptance rate: 21.98%=71/323).
2. Gyuyeong Kim and Wonjun Lee, "In-Network Leaderless Replication for Distributed Data Stores," in *Proc. of 48th International Conference on Very Large Data Bases* **(VLDB 2022)**, Sydney, Australia, September 2022.
3. Gyuyeong Kim and Wonjun Lee, "Protocol-Independent Service Queue Isolation for Multi-Queue Data Centers," in *Proc. of the 40th IEEE International Conference on Distributed Computing Systems* **(ICDCS 2020)**, Singapore, December 2020. (Acceptance rate: 17.98%=105/508)
4. Gyuyeong Kim and Wonjun Lee, "Service Function Chaining on Programmable Data Plane," in *Proc. of the 5th International Conference On Consumer Electronics Asia* (ICCE-Asia 2020), Busan, South Korea, November 2020.
5. Gyuyeong Kim and Wonjun Lee, "Stable Matching with Ties for Cloud-assisted Smart TV Services," in *Proc. of the 14th IEEE International Conference on Consumer Electronics* (ICCE 2014), Las Vegas, NV, January 2014.

6. Gyuyeong Kim and Wonjun Lee, "Cannot Take My Allocation: Enforcing Fairness by Considering Demand and Payment in Clouds," in *Proc. of the 4th International Conference on Network of the Future* (NoF 2013), Pohang, South Korea, October 2013.
7. Gyuyeong Kim, Hoorin Park, Jieun Yu, and Wonjun Lee, "Virtual Machines Placement for Network Isolation in Clouds," in *Proc. of 2012 Research in Applied Computation Symposium* (RACS 2012), San Antonio, TX, October 2012.

JOURNAL PUBLICATIONS

1. Gyuyeong Kim, "Switch-based Quorum Coordination for Low Tail Latency in Replicated Storage," *IEICE Transactions on Information and Systems*, Vol. E106-D, No. 11, pp. 1922-1925, November 2023.
2. Gyuyeong Kim, "Holistic In-Network Acceleration for Heavy-tailed Storage Workloads," *IEEE Access*, Vol. 11, pp. 77416-77428, July 2023.
3. Gyuyeong Kim and Wonjun Lee, "DynaQ: Enabling Protocol-Independent Service Queue Isolation in Cloud Data Centers," *IEEE Transactions on Cloud Computing* **(TCC)**, Vol. 11, No. 1, pp. 704-715, January/March 2023.
4. Gyuyeong Kim and Wonjun Lee, "LossPass: Absorbing Microbursts by Packet Eviction for Data Center Networks," *IEEE Transactions on Cloud Computing* **(TCC)**, Vol. 10, No. 4, pp. 2717-2728, October/December 2022.
5. Gyuyeong Kim and Wonjun Lee, "Network Policy Enforcement with Commodity Multiqueue NICs for Multi-Tenant Data Centers," *IEEE Internet of Things Journal* **(IoTJ)**, Vol. 9, No. 8, pp. 6252-6263, April 2022.
6. Gyuyeong Kim and Wonjun Lee, "Enabling Service Queue Isolation in Multi-Tenant Data Centers," *IEEE Communications Letters* **(CL)**, Vol. 23, No. 11, pp. 1949-1952, November 2019.
7. Gyuyeong Kim and Wonjun Lee, "Absorbing Microbursts without Headroom for Data Center Networks," *IEEE Communications Letters* **(CL)**, Vol. 23, No. 5, pp. 806-809, May 2019.
8. Gyuyeong Kim and Wonjun Lee, "Tardy Flow Scheduling in Data Center Networks," *IEICE Transactions on Information and Systems*, Vol. E99-D, No. 9, pp. 2400-2403, September 2016.

PATENTS

1. Wonjun Lee and Gyuyeong Kim, "System For Highly Available Service Function Chaining on Programmable Switches (고가용성을 지원하는 프로그래머블 스위치 상에서의 서비스 기능 체이닝 시스템)," *Korean Patent Registration No. 10-2416153*, June 14, 2022; *Application No. 10-2020-0141500*, October 28, 2020.
2. Wonjun Lee and Gyuyeong Kim, "Method for Managing Multiple Queues in Switch Ports, Recording Medium and Device for Performing the Method (스위치 포트 내 다중 대기열 관리 방법, 이를 수행하기 위한 기록 매체 및 장치)," *Korean Patent Registration No. 10-2291987*, August 13, 2021; *Application No. 10-2019-0135566*, October 29, 2019.
3. Wonjun Lee and Gyuyeong Kim, "Method to Enforce Network Policy with Commodity Multiqueue, Recording Medium and Device for Performing the Method (다중 큐를 지원하는 네트워크 정책 적용 방법, 이를 수행하기 위한 기록 매체 및 장치)," *Korean Patent Registration No. 10-2179769*, November 11, 2020; *Application No. 10-2019-0136628*, October 30, 2019.
4. Wonjun Lee and Gyuyeong Kim, "Method for Data Processing (데이터 처리 방법)," *Korean Patent Registration No. 10-2064679*, January 3, 2020; *Application No. 10-2018-0086807*, July 25, 2018.
5. Wonjun Lee and Gyuyeong Kim, "Method and Device for Scheduling Flow of Packet for Reducing Delay Time Due To Retransmit of Packet (패킷의 재전송으로 인한 지연을 단축시키기 위한 패킷의 플로우 스케줄링 방법 및 장치)," *Korean Patent Registration No. 10-1841143*, March 16, 2018; *Application No. 10-2016-0067533*, May 31, 2016.

EDUCATIONAL SERVICES

Department Chair, Dept. of Computer Engineering, SSWU	March 2023 - Present
Appointed Admissions Officer, SSWU	June 2022 - Present
Lecturer for High School Credit System Support Program, SSWU	Sept. 2022 - Present
Advisor for KOREN NET Challenge Camp Season 9 (Team: Masking)	Jun. 2022 - Dec. 2022
Lecturer for SW021 Program, University Employment Plus Center, SSWU	Aug. 2022

PROFESSIONAL ACTIVITIES

Reviewer

IEEE/ACM Transactions on Networking (ToN)
IEEE Transactions on Cloud Computing (TCC)
IEEE Communications Letters (CL)
IEEE Networking Letters
IEEE Open Journal of the Communications Society
IEEE Internet of Things Journal
ETRI Journal
IEEE International Conference on Computer Communications (INFOCOM), 2020
IEEE International Conference on Distributed Computing Systems (ICDCS), 2019

INVITED TALKS

NetClone: Fast, Scalable, and Dynamic Request Cloning for Microsecond-Scale RPCs
KIISE Korea Software Congress 2023, Busan, South Korea — Dec. 21, 2023

Network Switches as Domain-Specific Hardware for Distributed Storage
The 14th International Conference on ICT Convergence, Jeju, South Korea — Oct. 13, 2023

In-Network Acceleration for Modern Data Center Systems
Ulsan National Institute of Science and Technology (UNIST), Ulsan, South Korea — Sept. 6, 2023

P4 Switch Architecture
SDN/NFV Forum P4 WG 2020 1st Meetup, Seoul, South Korea — Dec. 14, 2020

Tutorial: P4 SFC
SDN/NFV Forum P4 WG 2019 1st Meetup, Seoul, South Korea — Apr. 19, 2019

Service Function Chaining in P4-enabled Programmable Switches
SDN/NFV Forum P4 WG 2018 2nd Meetup, Seoul, South Korea — Oct. 12, 2018

Controller-independent Loss-aware Low Latency State Migration in Network Functions Virtualization
ONOS-P4 Brigade Work Days 2017, Seoul, South Korea — Sept. 19, 2017

HONORS AND AWARDS

IEEE Seoul Section International Student Paper Contest Bronze Paper Award	2014
Global Ph.D. Fellowship (GPF), NRF, South Korea	2012 - 2014
ACM-ICPC Seoul Regional Contest 10th Place	2009
National Collegiate Programming Contest Silver Prize, South Korea	2009
Sun Microsystems JavaFX Software Contest 3rd Prize	2009
ACM-ICPC Seoul Regional Contest Honorable Mentions	2008

REMARKS

Co-founder and President, KLUE (Korea University Lecture Evaluation) — Apr. 2010 - Feb. 2015
- KLUE (http://klue.kr) is a lecture rating service for Korea University whose sources are based on student feedback. KLUE has 500K+ evaluation data with 50K+ members now.
- Brief history: In 2010, upon the request of the 43rd Student Council, 3 (D. Kim, I. Seo, and me) of the KWEB student club began a development project. Our team was in charge of the whole planning

and development. The service was run by the student council as planned. However, in November 2010, the student council violated the privacy policy and the service is closed. In January 2011, I revived the service and became president. The 44th, 45th, and 46th Student Councils generously provided support for server costs until we achieved financial self-sufficiency. The experience in developing and operating KLUE led me to explore networked systems during my Ph.D.